

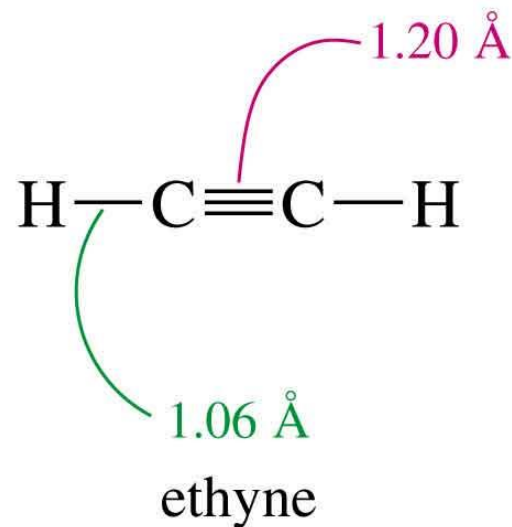
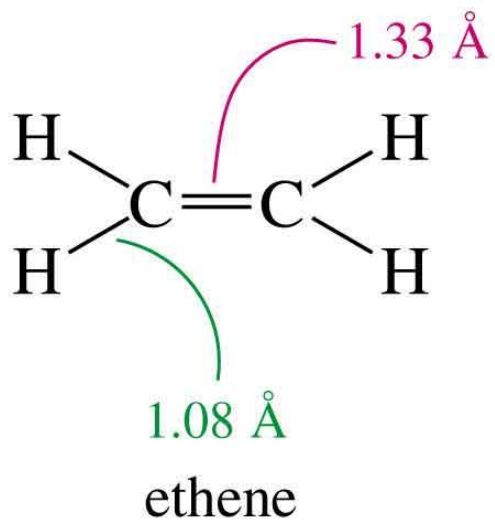
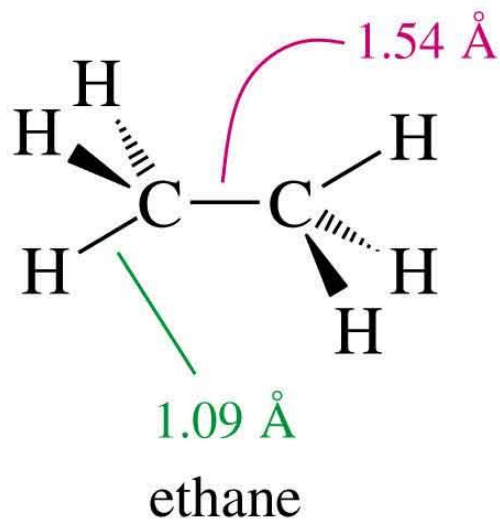
Alkynes

Alkynes

- Alkynes contain one or more triple bond.
- General formula is C_nH_{2n-2}
- Two elements of unsaturation for each triple bond.
- Some reactions are like alkenes: addition and oxidation.
- Nonpolar, insoluble in water.
- Soluble in most organic solvents.
- Boiling points similar to alkane of same size.
- Less dense than water.
- Up to 4 carbons, gas at room temperature.

Bond Lengths

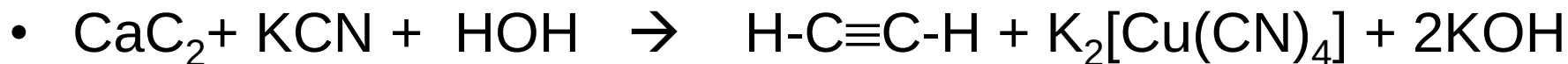
- More *s* (than *p*) character, so shorter length.
- Three bonding overlaps, so shorter.



Bond angle is 180° , so linear geometry.

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Preparation of Alkynes: Acetylene



- Acetylene can be prepared from calcium carbide and water.

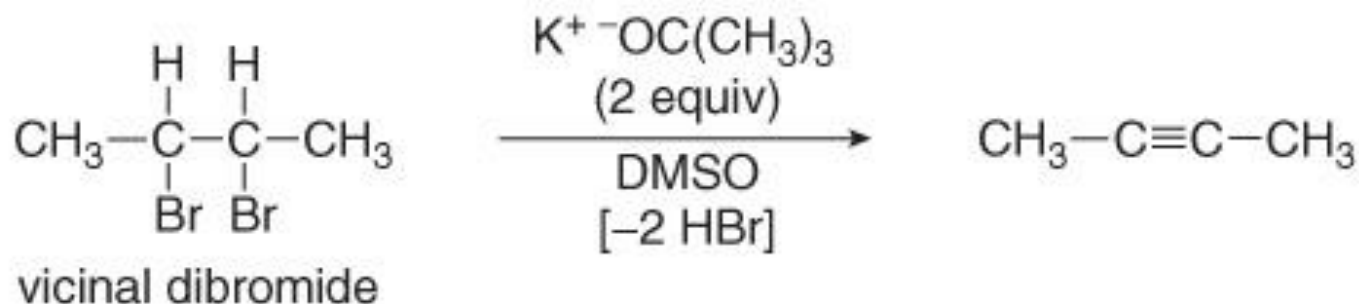
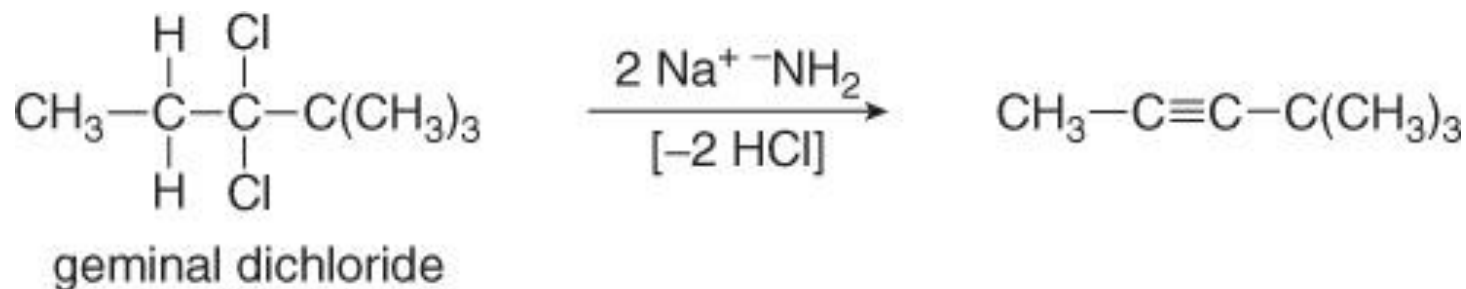


- Calcium carbide is prepared by heating coke (Raw Carbon) and calcium oxide in an electric furnace ($\sim 2500^\circ$).

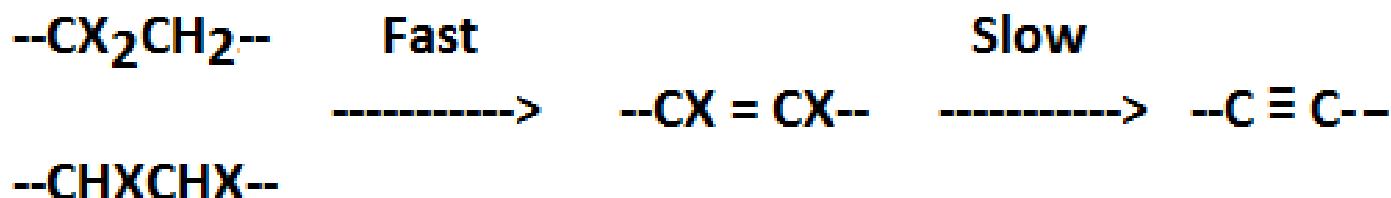


Preparation of Alkynes by Dehydrohalogenation

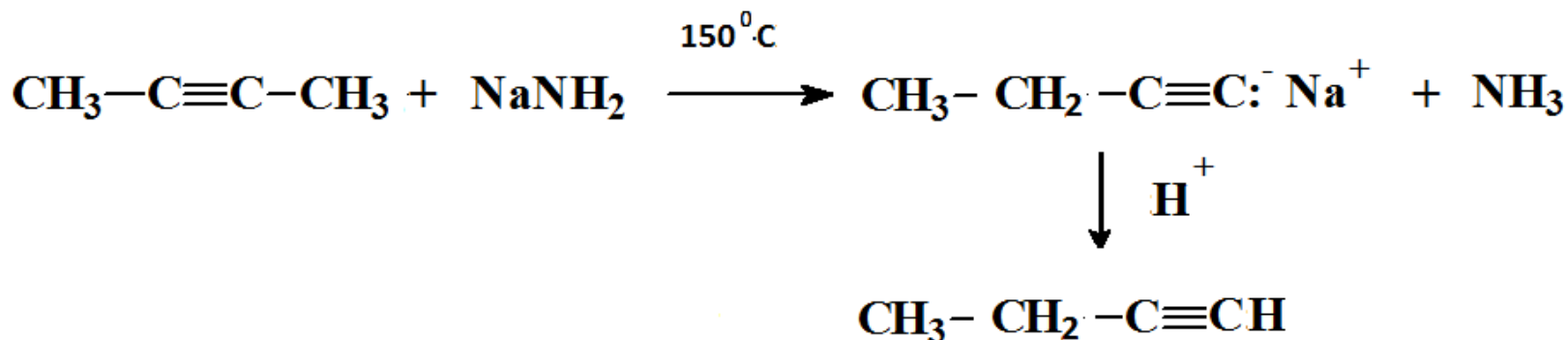
- alkynes are prepared by elimination reactions. A strong base removes two equivalents of HX from a vicinal or geminal dihalide to yield an alkyne through two successive E2 elimination reactions.



- Dehydrohalogenation proceeds in two stages. The removal of 2nd HX molecule is difficulty than removal of 1st HX, requires very strong base and high temperatures.



- Internal alkynes may be converted into terminal alkynes by NaNH₂ at 150°C (H⁺ can be removed from a terminal alkyne by sodium amide, NaNH₂).



Reactions of Alkynes

1. Addition reactions
2. Substitution reactions
3. Oxidation reactions

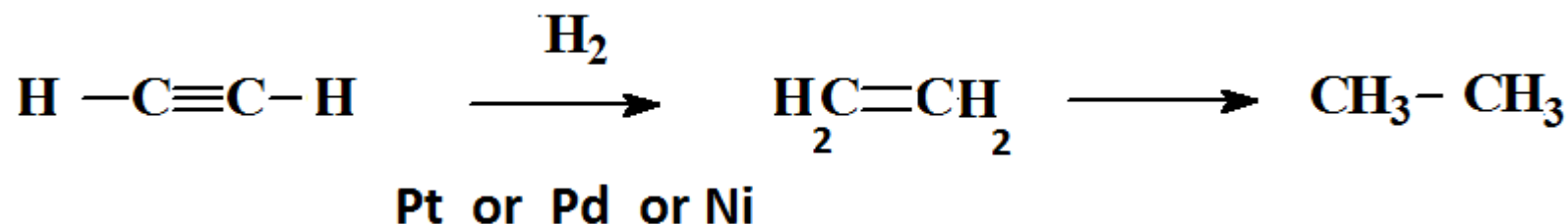
1. Addition reactions

As a general rule, alkynes react with electrophilic reagents in Manner similar to that of alkenes

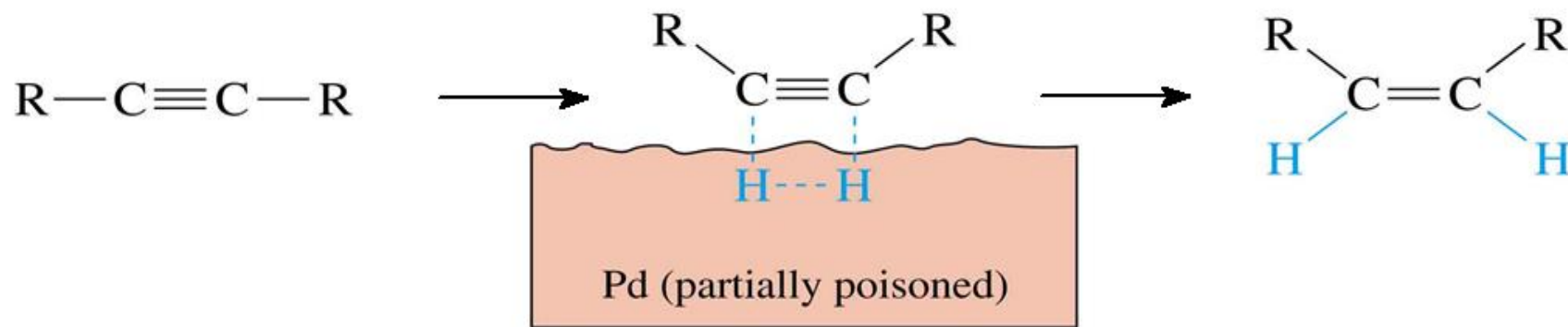
Addition of Hydrogen (Hydrogenation)

Three reactions:

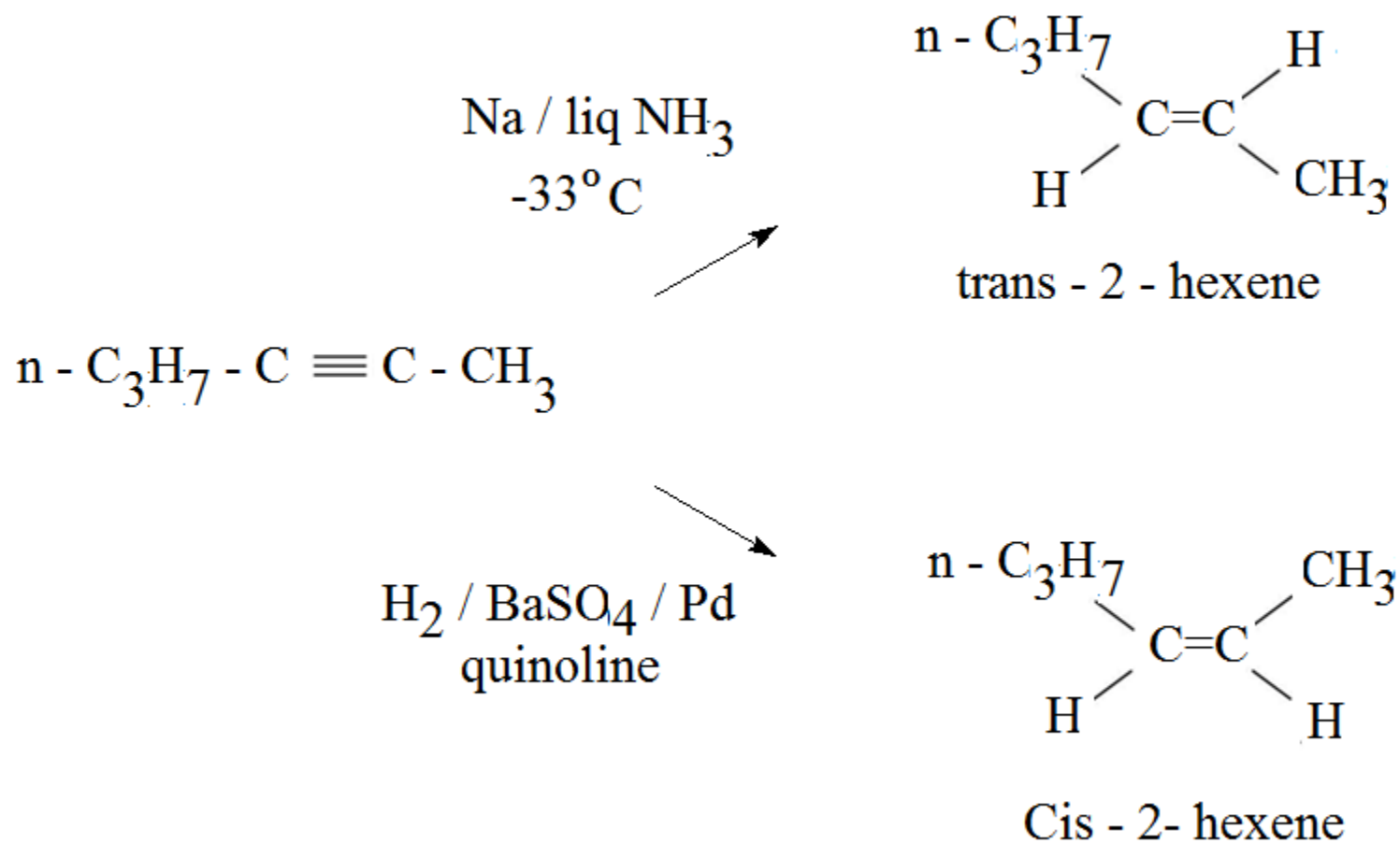
- Add lots of H_2 with metal catalyst (Pd, Pt, or Ni) to reduce alkyne to alkane, completely saturated.



- Use a special catalyst, Lindlar's catalyst to convert an alkyne to a **cis**-alkene (Powdered BaSO_4 coated with Pd, poisoned with quinoline).

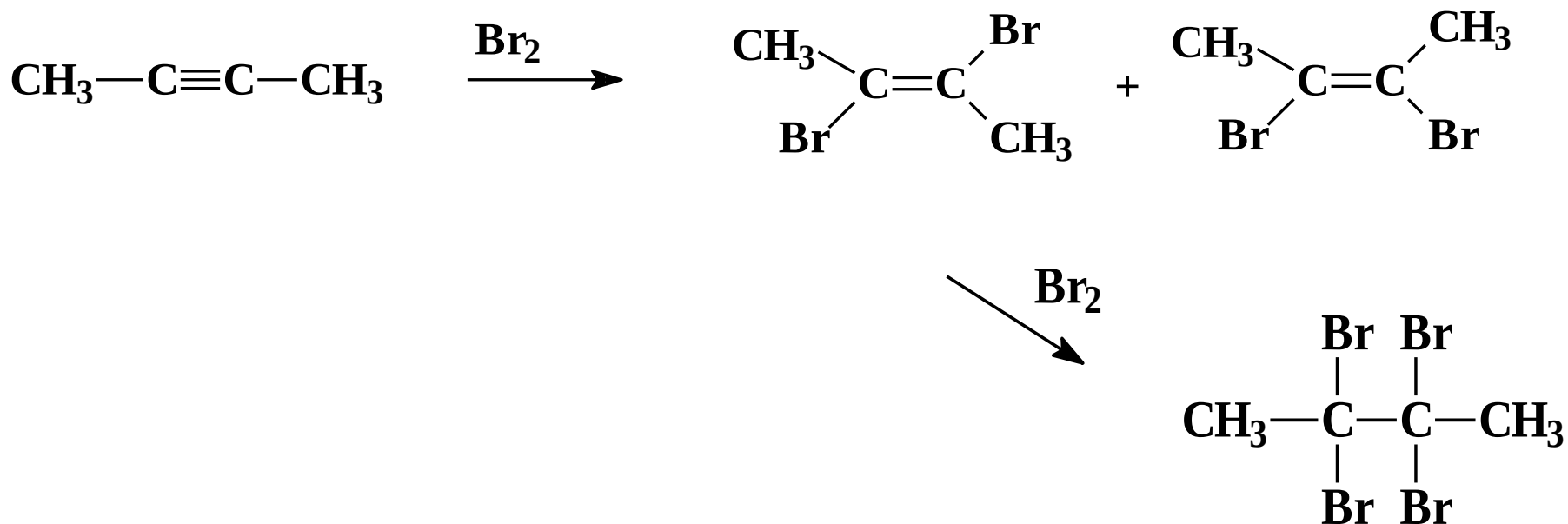


- React the alkyne with sodium in liquid ammonia to form a trans-alkene (Free radical reaction)



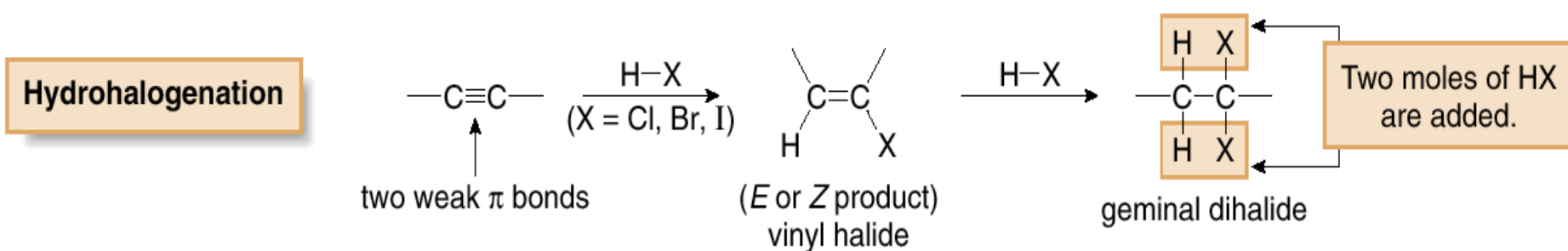
Addition of Halogens

- Cl_2 and Br_2 add to alkynes to form vinyl dihalides.
- May add syn or anti, so product is mixture of **cis** and **trans** isomers.
- Difficult to stop the reaction at dihalide.



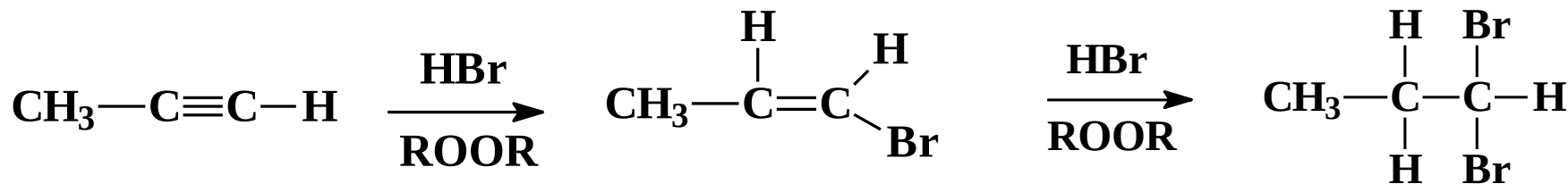
Addition of HX

- HCl, HBr, and HI add to alkynes to form vinyl halides.
- For terminal alkynes, Markovnikov product is formed.
- If two moles of HX is added, product is a geminal dihalide.



HBr with Peroxides

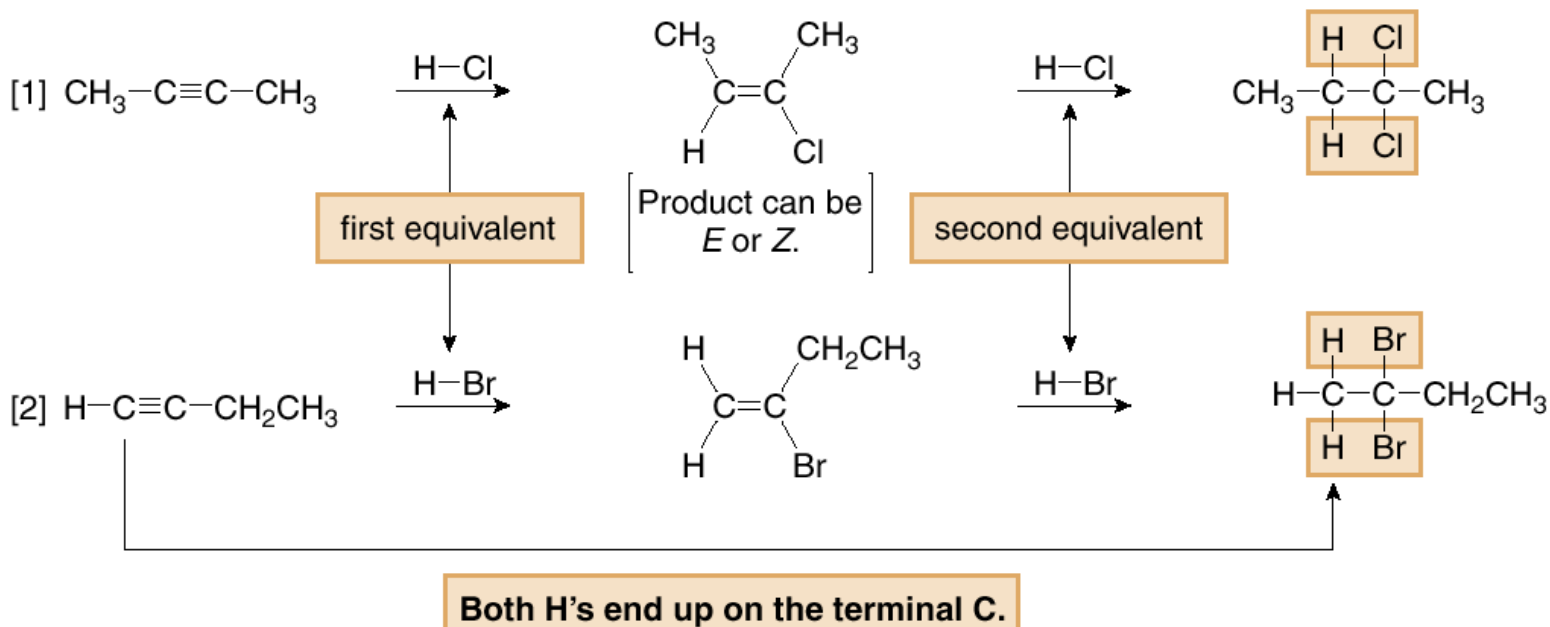
- Anti-Markovnikov product is formed with a terminal alkyne.



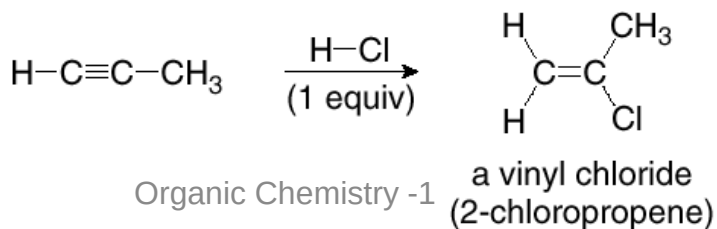
Hydrohalogenation: Electrophilic Addition of HX

- With two equivalents of HX, both H atoms bond to the *same* carbon.
- With a terminal alkyne, both H atoms bond to the *terminal* carbon; that is, the hydrohalogenation of alkynes follows Markovnikov's rule.

Examples



- With only one equivalent of HX, the reaction stops with formation of the vinyl halide.

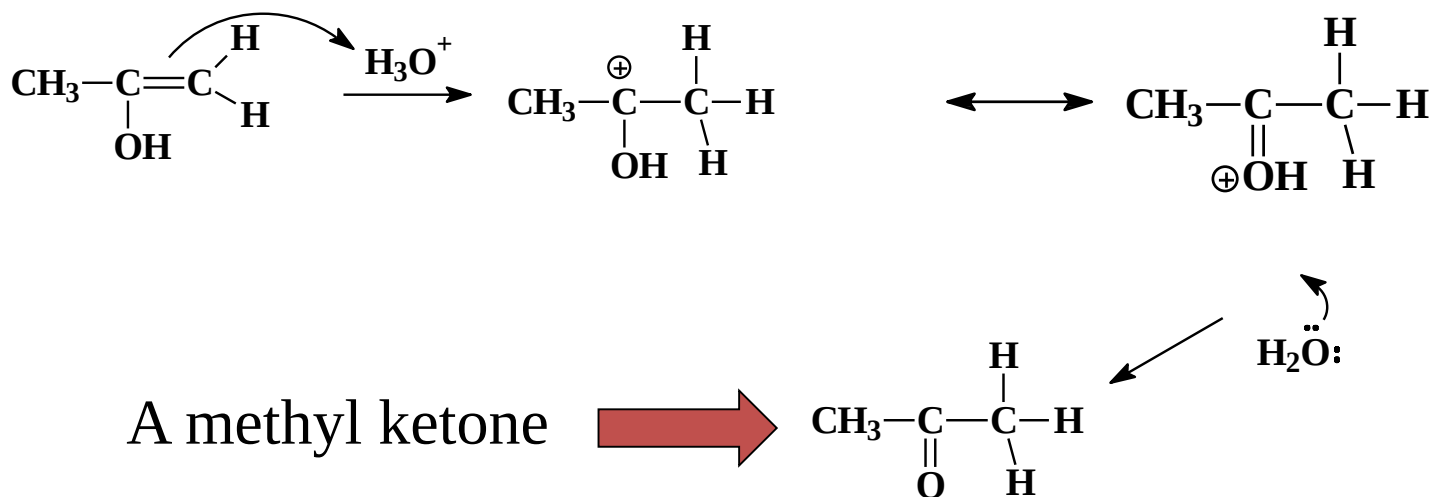


Hydration of Alkynes

- Mercuric sulfate in aqueous sulfuric acid adds H-OH to one pi bond with a Markovnikov orientation, forming a vinyl alcohol (enol) that rearranges to a ketone.
- Hydroboration-oxidation adds H-OH with an anti-Markovnikov orientation, and rearranges to an aldehyde.

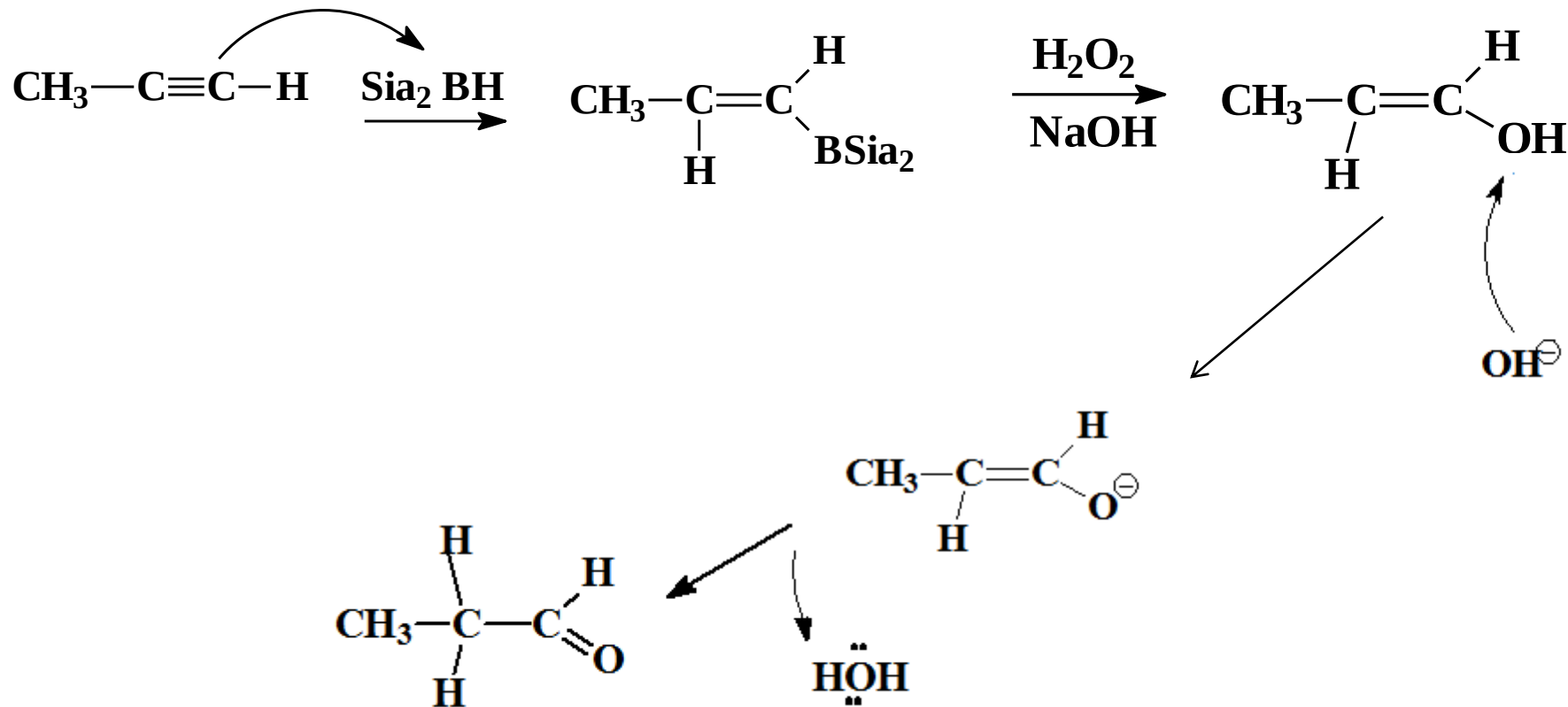
Enol to Keto (in Acid)

- Add H^+ to the $\text{C}=\text{C}$ double bond.
- Remove H^+ from OH of the enol.



Hydroboration - Oxidation

- B and H of Sia_2BH (Disiamyl borane) add across the triple bond.
- Oxidation with basic H_2O_2 gives the enol.
- H^+ is removed from OH of the enol.
- Then water gives H^+ to the adjacent carbon convert to aldehyde or ketone.

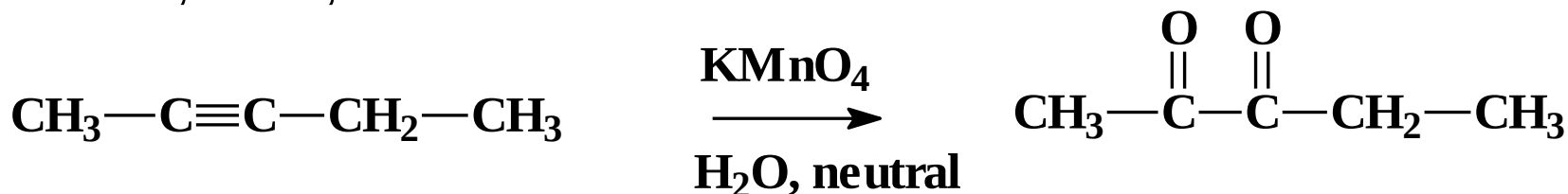


Oxidation of Alkynes

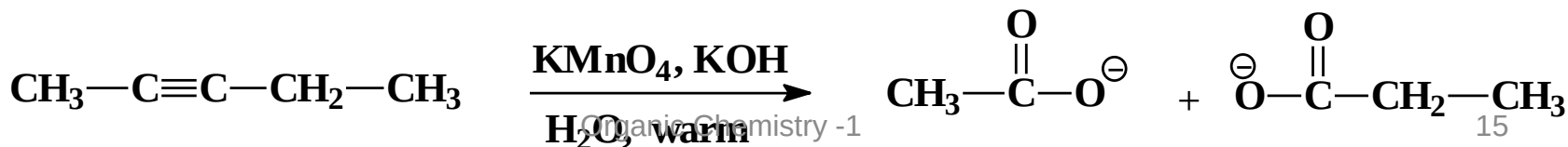
- Similar to oxidation of alkenes.
- Dilute, neutral solution of KMnO_4 oxidizes alkynes to a diketone.
- Warm, basic KMnO_4 cleaves the triple bond.
- Ozonolysis, followed by hydrolysis, cleaves the triple bond.

Reaction with KMnO_4

Mild conditions, dilute, neutral

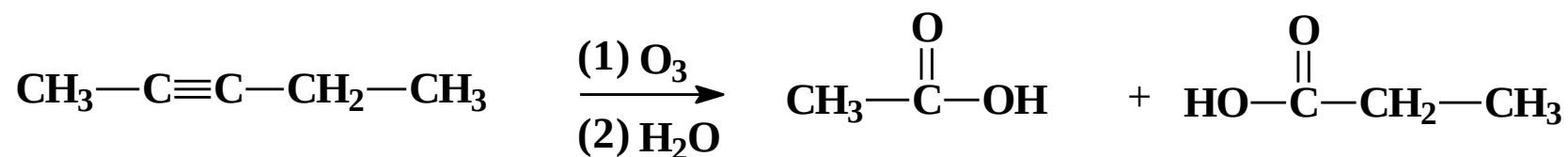


Harsher conditions, warm, basic



Ozonolysis

- Ozonolysis of alkynes produces carboxylic acids (Alkenes gave aldehydes and ketones)



- Used to find location of triple bond in an unknown compound.

Four addition reactions of 1-butyne

